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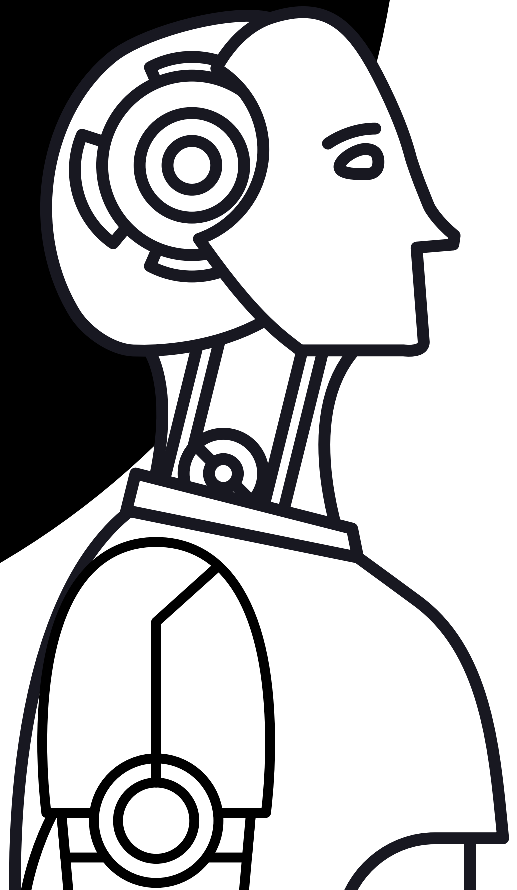
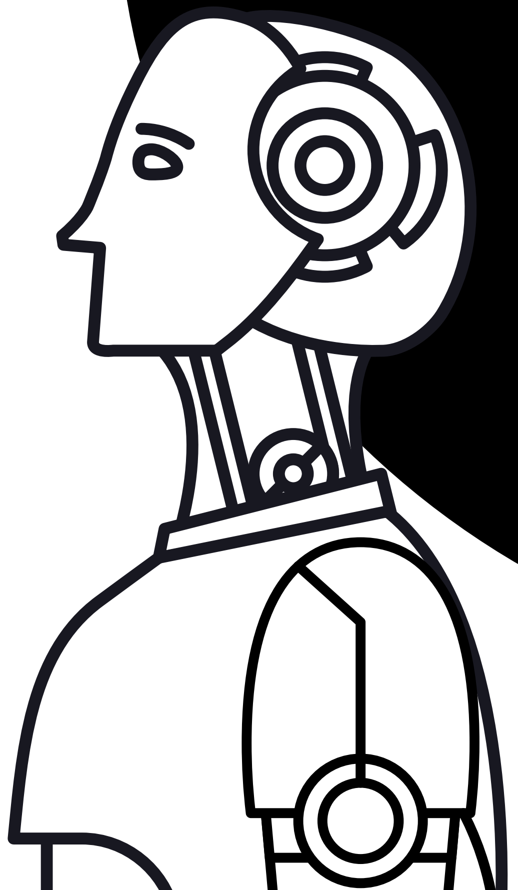
ROBUST LIDAR-BASED NAVIGATION WITH REINFORCEMENT LEARNING IN WEBOTS

INTRODUCTION TO INTELLIGENT ROBOTICS PRESENTATION



This project studies how different RL navigation models behave under sensor degradation, map changes, and different training strategies.



PROJECT GOAL

- Implement and evaluate DRL-based autonomous navigation architectures in Webots.
- Investigate policy robustness against sensor degradation:
  - Sensitivity analysis to LiDAR noise levels.
  - Performance impact of LiDAR resolution (ray density) reduction.
- Analyze spatial generalization and performance:
  - Policy adaptation across structural map variations.
  - Comparative benchmarking: Continuous PPO vs. Discrete DQN in complex hallway layouts.
- Quantitative Performance Metrics:
  - Success rate, collision rate, timeout rate and average steps to goal.

MOTIVATION & CHALLENGES

- The Sim-to-Real Gap: DRL policies often overfit to pristine simulated data.
- Sensor Reliability: High-fidelity LiDAR data is rarely available in real-world scenarios.
- Generalization vs. Adaptation: Can agents trained in simple environments navigate complex, dynamic structures?
- Control Paradigms: Comparing continuous (PPO) vs. discrete (DQN) strategies under sensor constraints.

METHODOLOGY: THREE-PHASE EVALUATION

- Phase 1: Noise Threshold Determination
  - Comparison of baseline paradigms (Model A: Ideal vs. Model B: Over-Noised).
  - Identification of the optimal training noise threshold (0.05).
- Phase 2: Resolution and Ray-Density Analysis
  - Evaluation of CNN feature extraction with 360-ray (Model C) vs. 90-ray (Model D) LiDAR configurations.
  - Selection of Model C as the foundational baseline.
- Phase 3: Spatial Generalization and Benchmarking
  - Evaluation in smart-wheelchairs-complex.wbt (complex architectural challenges).
  - PPO: Fine-tuning (1M additional timesteps).
  - DQN: Training from scratch.
  - Reactive Testing: Dynamic obstacle stress test.

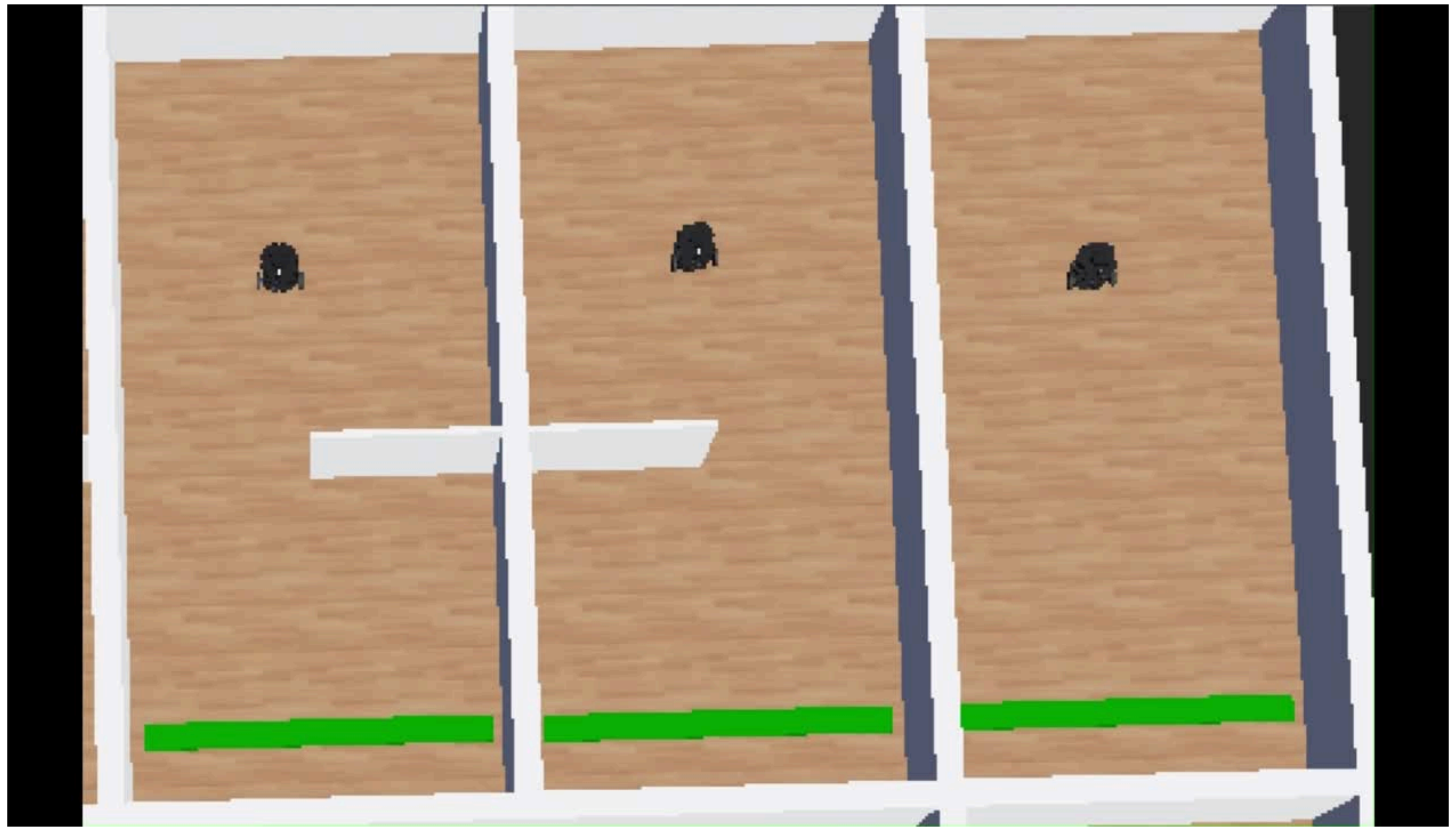
NOISE AND LIDAR SENSITIVITY

Model A: trained with 0 noise, 360 rays.
Model B: trained with 0.5 noise, 36 rays.
Tested across:

- noise = 0.1, 0.2, 0.5;
- rays = 36, 90, 180, 360.

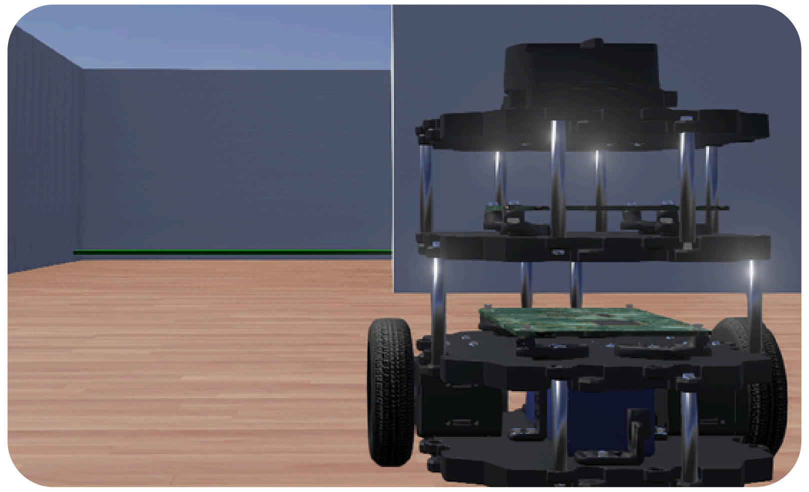
Main findings:

- Noise had a much stronger impact than ray reduction;
- Model A stayed robust under ray reduction;
- Model B did not show the expected robustness;
- Chosen standard noise for the next phases: 0.05.



Model A (baseline)

CHOOSING THE BEST RAY CONFIGURATION



Model C trained with 360 rays.
Model D trained with 90 rays.
Both tested with:

- 360, 180, 90, 36 rays.

Main result:

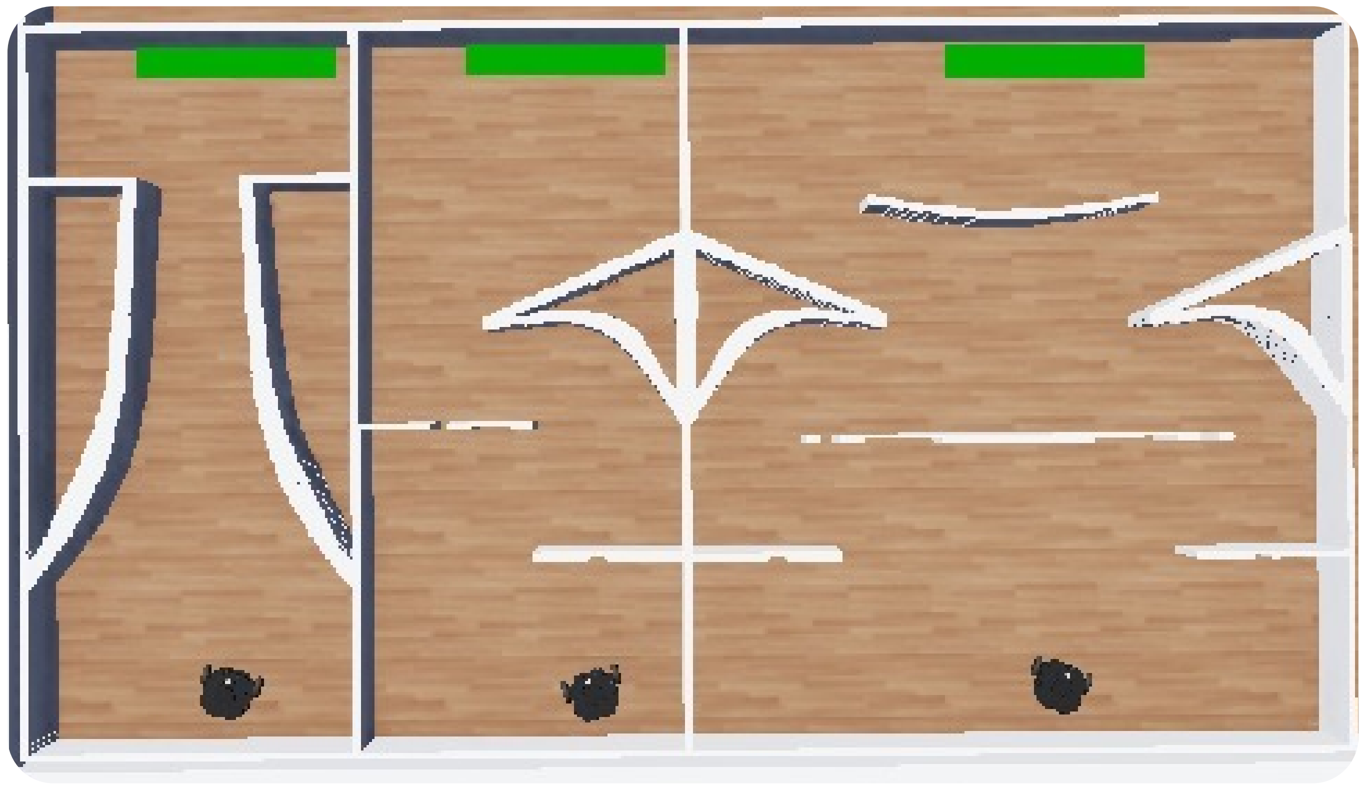
- Both reached 100% success;
- Therefore, ray count did not significantly affect performance in the original map.

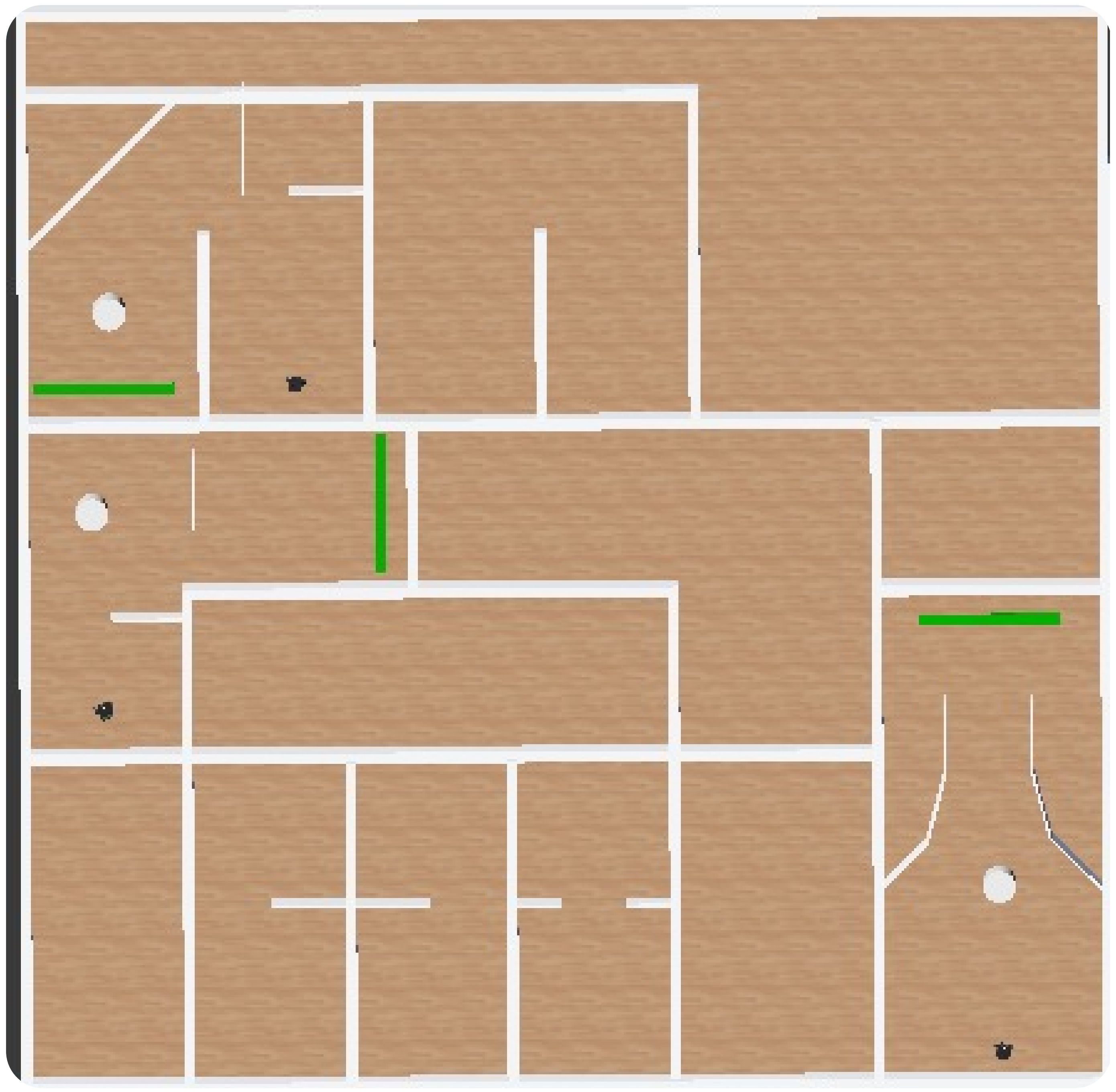
Since both models performed perfectly, Model C was kept as the base for Phase 3, mainly due to compatibility issues with Model D.

NEW MAP FOR GENERALIZATION

A first completely new map was created.
Robots could not move properly and showed unstable oscillatory behavior.
Therefore, the original map was modified instead:

- Previously unused corridors were activated;
- Additional obstacles were introduced;
- The layout became more complete and challenging.





GENERALIZATION OF THE ORIGINAL MODEL C

Model C was then tested on the modified map.

01

RESULTS

- Overall success rate: 8.24%;
- Robot 0: 43.75%;
- Robot 1: 0%;
- Robot 2: 0%.

02

INTERPRETATION

The original Model C showed very limited generalization.
Only one starting corridor retained partial success
The remaining two corridors failed completely.

This showed that the original policy had learned useful navigation behavior, but that it was strongly tied to the structure of the original map.

PPO FINE-TUNING ON THE NEW MAP

Model C was then retrained on the modified map

01

RESULTS

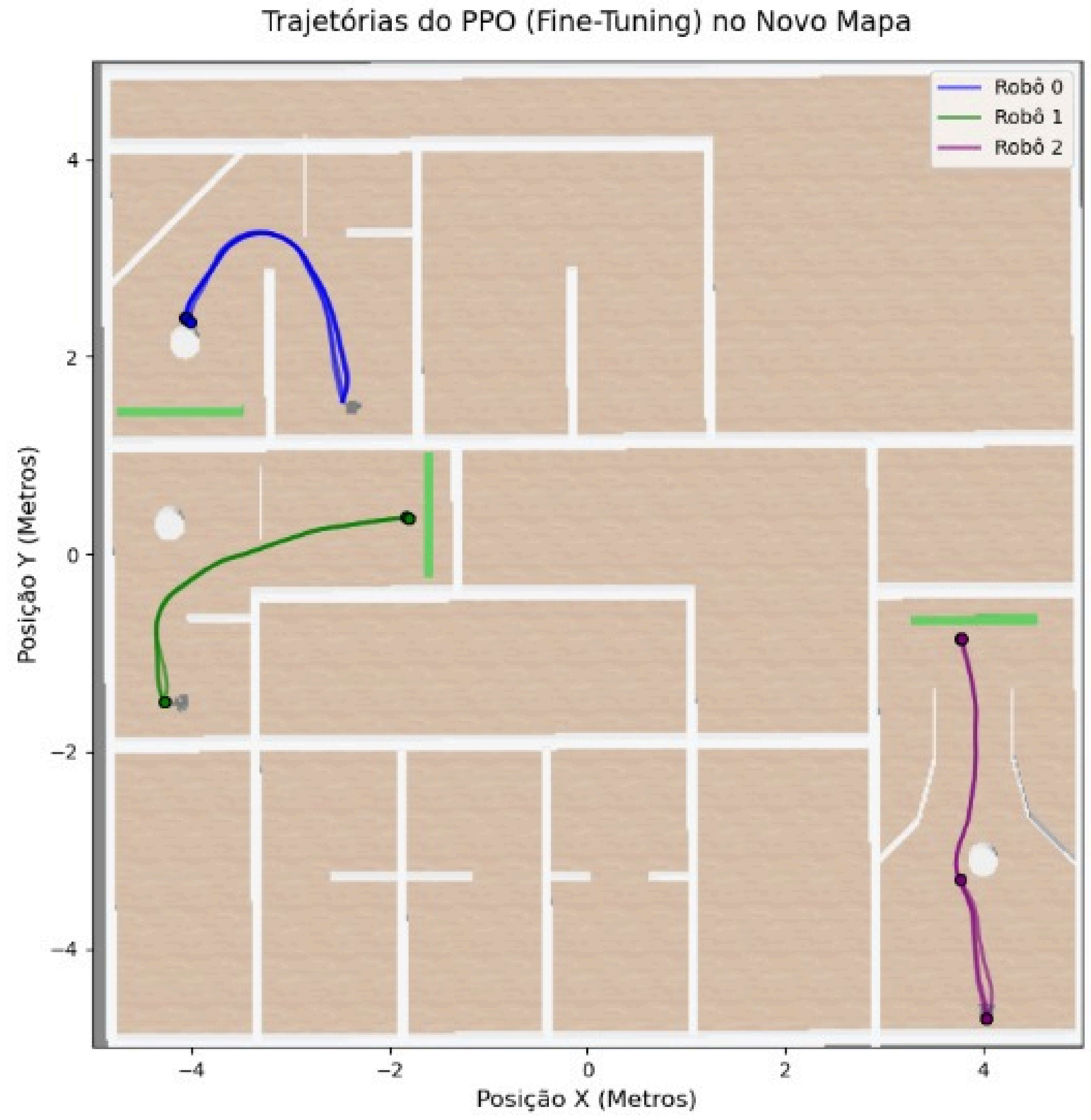
- Overall success rate: 46%;
- Robot 0: 0%;
- Robot 1: 100%;
- Robot 2: 47.37%.

02

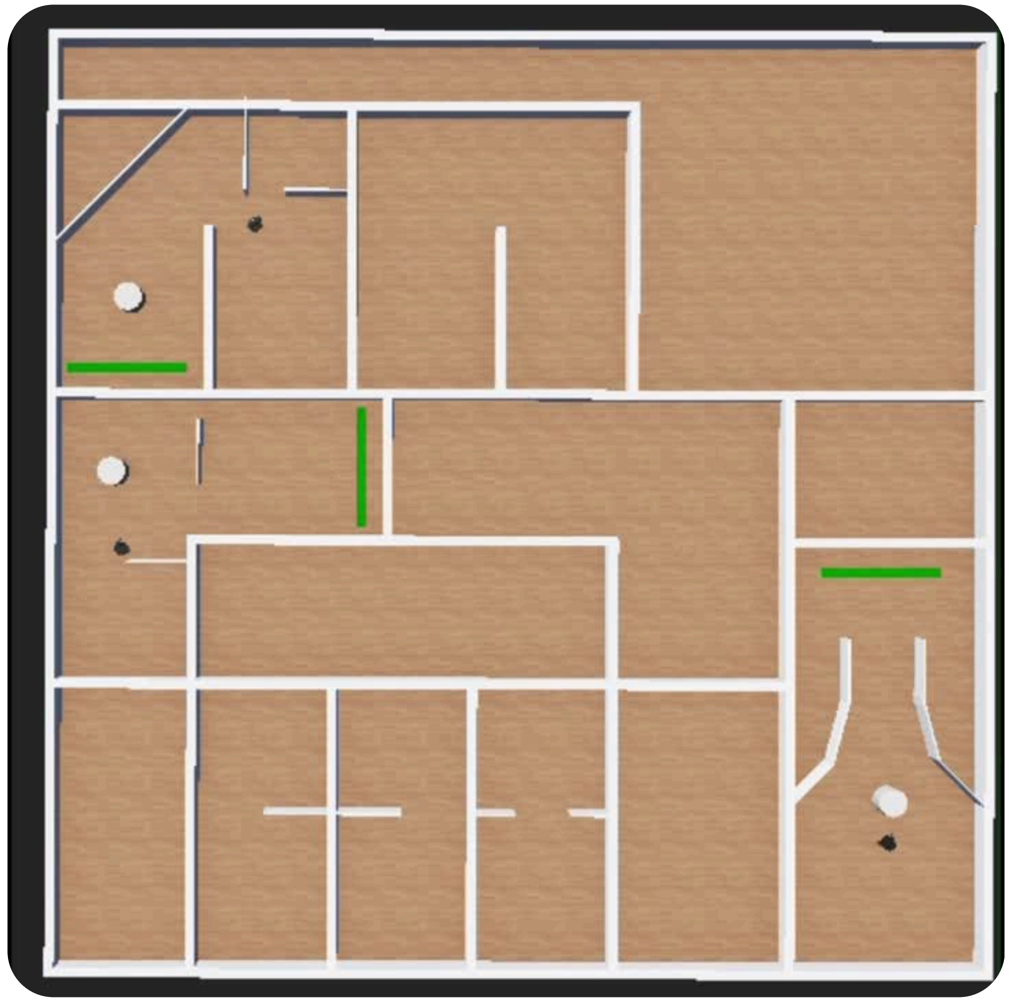
INTERPRETATION

Retraining clearly improved performance compared to the original Model C.
However, performance remained highly corridor-dependent.
One corridor became fully solved, one partially solved, and one still failed completely.

PPO adapted to the new environment, but the learned policy remained specialized rather than fully general.



DQN ON THE NEW MAP



01

RESULTS

- Overall success rate: 65%;
- Robot 0: 0%;
- Robot 1: 100%;
- Robot 2: 100%.

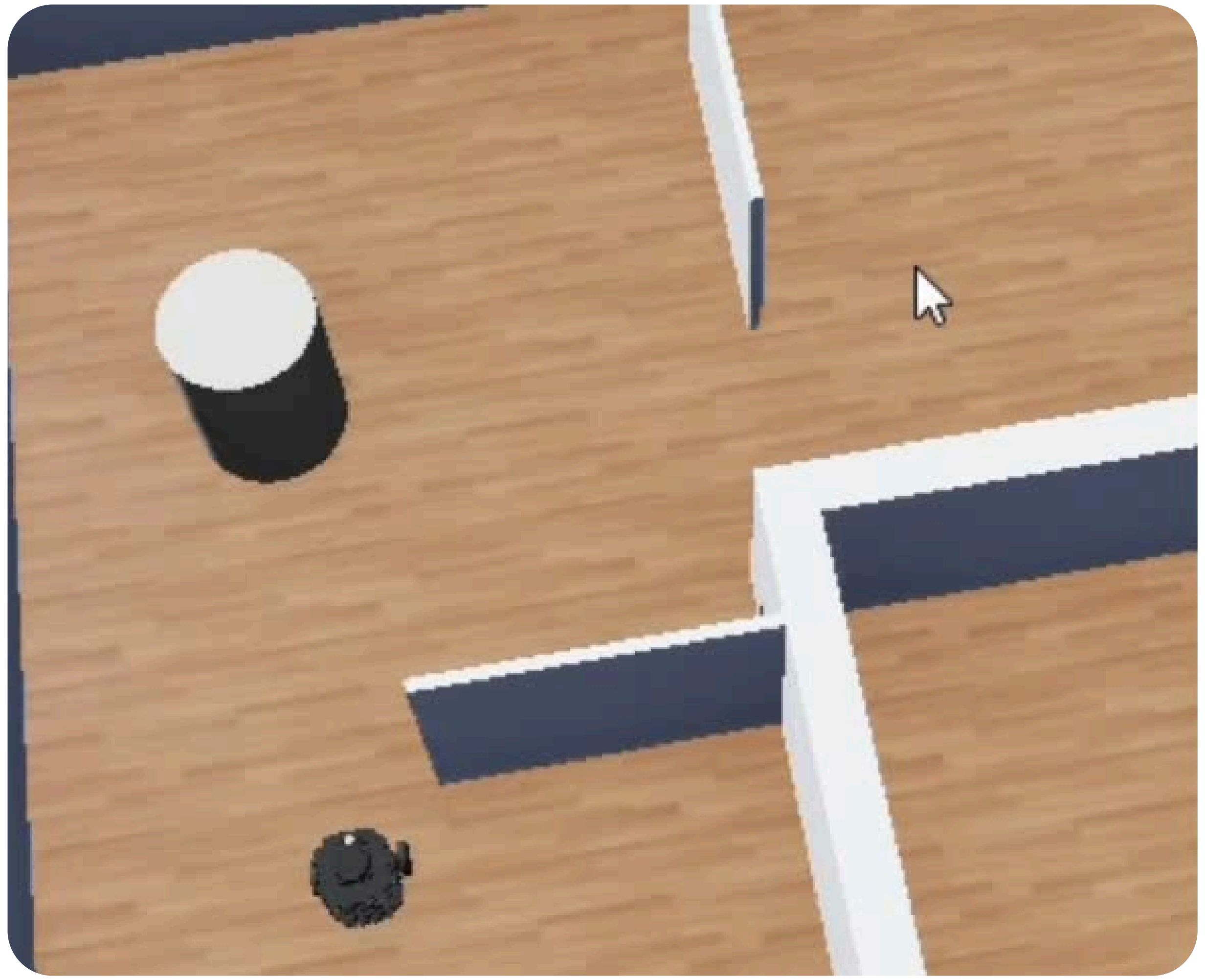
02

INTERPRETATION

DQN achieved the highest aggregate success rate.
However, it still failed completely in one corridor.
This suggests strong specialization to specific initial conditions.

Although DQN outperformed PPO in aggregate success rate on the new map, its complete failure in one corridor shows that the policy still lacked uniform generalization across all scenarios.

REACTIVE NAVIGATION: DYNAMIC OBSTACLE



01

RESULTS

- Success rate: 0% (in intersection scenarios);
- Behavior: Failure to adjust trajectory;
- Interaction: Consistent collisions with moving pillar.

02

INTERPRETATION

Lack of Temporal Awareness: The LiDAR-only CNN architecture treats moving objects as static geometry.
Prediction Failure: The agent fails to extrapolate velocity vectors of dynamic obstacles.

While these results highlight a fundamental limitation in temporal awareness, they represent a preliminary investigation intended to define future requirements for recurrent neural network integration or frame-stacking techniques.

FINAL CONCLUSIONS

The main challenge was not simply sensing less, but learning navigation policies that remained reliable when the environment changed.

01

NOISE SENSITIVITY

Noise was more harmful than reducing LiDAR rays
A noise level of 0.05 was selected as the standard setting.

02

LIDAR RESOLUTION EFFECTS

In the original map, LiDAR ray count had little effect on performance.

03

CROSS-MAP GENERALIZATION

- The original Model C generalized poorly to the new map (8.24%);
- Retraining PPO improved the result to 46%;
- DQN reached the best aggregate result on the new map (65%);
- However, both PPO and DQN still showed corridor-specific weaknesses;
- Tests involving moving obstacles revealed a fundamental lack of temporal awareness.

PROJECT CONTRIBUTIONS

- Equal Contribution: 33.3% each

- Sebastian:
  - Spearheaded Phase 1 research, noise threshold determination, and baseline paradigm establishment.
- Martim:
  - Led Phase 2 development, including LiDAR resolution analysis.
- Maria Inês Rocha:
  - Coordinated Phase 3 implementation, spatial generalization testing, and algorithmic benchmarking (PPO vs. DQN).

- Collaborative Efforts: All team members actively participated in the integration of the Webots/Python architecture, shared debugging of the ZMQ communication protocol, and contributed to the overall analysis of the final results.